Full Stack Development — Mini E-Commerce Project

Lab Sheet 3: Rebuilding with React (Components, Props, useState)

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Roll No. – 68

Main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";

import "./App.css";
import App from "./App.jsx";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <App />
  </StrictMode>
);
```

Cart.jsx

```
function Cart({ cart, removeFromCart }) {
  const grandTotal = cart.reduce(
    (total, item) => total + item.product.price * item.quantity,
    0
  );

  return (
    <section className="cart-section">
      <h2>Your Cart</h2>

      {cart.length === 0 ? (
        <p className="empty-cart">Your cart is empty</p>
      ) : (
        <>
          <div className="cart-items">
            {cart.map((item) => (
              <div className="cart-item" key={item.product.id}>
                <div>
```

```
              <h3>{item.product.name}</h3>
              <p>Price: ₹{item.product.price}</p>
              <p>Quantity: {item.quantity}</p>
              <p>
                Subtotal: ₹
                {item.product.price * item.quantity}
              </p>
            </div>
            <button
              className="remove-button"
              onClick={() => removeFromCart(item.product.id)}
            >
              Remove
            </button>
          </div>
        ))}
      </div>
      <div className="grand-total">
        <h3>Grand Total: ₹{grandTotal}</h3>
      </div>
    </>
  )}
 </section>
 );
}
export default Cart;
```

CheckoutForm.jsx

```
import { useState } from "react";

function CheckoutForm() {
 const [formData, setFormData] = useState({
  name: "",
  address: "",
  pincode: "",
  phone: "",
  payment: ""
 });

 const handleChange = (event) => {
  const { name, value } = event.target;

  setFormData({
   ...formData,
   [name]: value
  });
```

```
};

const handleSubmit = (event) => {
  event.preventDefault();

  alert("Order placed successfully!");

  console.log("Order Details:", formData);
};

return (
  <section className="checkout-section">
    <h2>Checkout</h2>

    <form onSubmit={handleSubmit}>
      <div className="form-group">
        <label>Name</label>

        <input
          type="text"
          name="name"
          value={formData.name}
          onChange={handleChange}
          required
        />
      </div>

      <div className="form-group">
        <label>Address</label>

        <textarea
          name="address"
          value={formData.address}
          onChange={handleChange}
          rows="4"
          required
        ></textarea>
      </div>

      <div className="form-group">
        <label>Pincode</label>

        <input
          type="text"
          name="pincode"
          value={formData.pincode}
```

```
    onChange={handleChange}
    pattern="[0-9]{6}"
    maxLength="6"
    required
   />
</div>

<div className="form-group">
  <label>Phone</label>

  <input
    type="tel"
    name="phone"
    value={formData.phone}
    onChange={handleChange}
    pattern="[0-9]{10}"
    maxLength="10"
    required
   />
</div>

<div className="payment-section">
  <h3>Payment Method</h3>

  <label>
    <input
      type="radio"
      name="payment"
      value="Cash on Delivery"
      checked={formData.payment === "Cash on Delivery"}
      onChange={handleChange}
      required
    />
    Cash on Delivery
  </label>

  <label>
    <input
      type="radio"
      name="payment"
      value="UPI"
      checked={formData.payment === "UPI"}
      onChange={handleChange}
    />
    UPI
  </label>
```

```
          <label>
            <input
              type="radio"
              name="payment"
              value="Credit/Debit Card"
              checked={formData.payment === "Credit/Debit Card"}
              onChange={handleChange}
            />
            Credit/Debit Card
          </label>
        </div>

        <button type="submit">
          Place Order
        </button>
      </form>
    </section>
  );
}

export default CheckoutForm;
```

ProductCart.jsx

```
function ProductCard({ product, addToCart }) {
  return (
    <div className="product-card">
      <img
        src={product.image}
        alt={product.name}
      />

      <h3>{product.name}</h3>

      <p className="price">₹{product.price}</p>

      <button onClick={() => addToCart(product)}>
        Add to Cart
      </button>
    </div>
  );
}
export default ProductCard;
```

ProductList.jsx

```
import products from "../products";
import ProductCard from "./ProductCard";

function ProductList({ addToCart }) {
  return (
    <section className="products-section">
      <h2>All Products</h2>

      <div className="product-grid">
        {products.map((product) => (
          <ProductCard
            key={product.id}
            product={product}
            addToCart={addToCart}
          />
        ))}
      </div>
    </section>
  );
}
export default ProductList;
```

App.jsx

```
import { useState } from "react";

import ProductList from "./components/ProductList";
import Cart from "./components/Cart";
import CheckoutForm from "./components/CheckoutForm";

function App() {
  const [cart, setCart] = useState([]);

  const [showCart, setShowCart] = useState(false);

  const addToCart = (product) => {
    setCart((previousCart) => {
      const existingProduct = previousCart.find(
        (item) => item.product.id === product.id
      );

      if (existingProduct) {
        return previousCart.map((item) =>
          item.product.id === product.id
            ? {
                ...item,
                quantity: item.quantity + 1
```

```
            }
          : item
      );
    }

    return [
      ...previousCart,
      {
        product: product,
        quantity: 1
      }
    ];
  });
};

const removeFromCart = (productId) => {
  setCart((previousCart) =>
  previousCart.filter(
      (item) => item.product.id !== productId
    )
  );
};

const cartItemCount = cart.reduce(
  (total, item) => total + item.quantity,
  0
);

return (
  <div className="app">

    <header className="navbar">
      <div className="logo">
        <h1>MyStore</h1>
      </div>

      <nav>
        <a href="#home">Home</a>

        <a href="#products">
          Products
        </a>

        <button
          className="cart-button"
          onClick={() => setShowCart(!showCart)}
        >
          🛒 Cart
          <span className="cart-badge">
            {cartItemCount}
```

```
            </span>
          </button>
        </nav>
      </header>

      <main>

        <section id="home" className="hero">
          <h2>Welcome to MyStore</h2>

          <p>
            Discover quality products at affordable prices.
          </p>

          <a href="#products">
            Shop Now
          </a>
        </section>

        <section id="products">
          <ProductList addToCart={addToCart} />
        </section>

        {showCart && (
          <Cart
            cart={cart}
            removeFromCart={removeFromCart}
          />
        )}

        <CheckoutForm />

      </main>

      <footer>
        <p>Contact: support@mystore.com</p>
        <p>Phone: +91 9898989898</p>
        <p>© 2026 MyStore. All Rights Reserved.</p>
      </footer>

    </div>
  );
}

export default App;
```

App.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

:root {
  --primary-color: #1e3a8a;
  --secondary-color: #2563eb;
  --accent-color: #f59e0b;
  --background-color: #f8fafc;
  --text-color: #1f2937;
}

body {
  font-family: Arial, Helvetica, sans-serif;
  background: var(--background-color);
  color: var(--text-color);
  line-height: 1.6;
}

html {
  scroll-behavior: smooth;
}

.navbar {
  background: var(--primary-color);
  color: white;
  padding: 20px 7%;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.logo h1 {
  font-size: 28px;
}

nav {
  display: flex;
  align-items: center;
  gap: 25px;
}

nav a {
  color: white;
  text-decoration: none;
  font-weight: bold;
```

```
}

nav a:hover {
  color: var(--accent-color);
}

.cart-button {
  background: var(--secondary-color);
  color: white;
  border: none;
  padding: 10px 18px;
  border-radius: 8px;
  cursor: pointer;
  font-size: 15px;
}

.cart-button:hover {
  background: var(--accent-color);
}

.cart-badge {
  background: var(--accent-color);
  color: black;
  border-radius: 50%;
  padding: 3px 8px;
  margin-left: 5px;
  font-weight: bold;
}

main {
  width: 90%;
  max-width: 1200px;
  margin: auto;
}

.hero {
  text-align: center;
  padding: 70px 20px;
}

.hero h2 {
  font-size: 36px;
  margin-bottom: 10px;
}

.hero p {
  font-size: 18px;
  margin-bottom: 25px;
}
```

```
.hero a {
  display: inline-block;
  background: var(--secondary-color);
  color: white;
  padding: 12px 25px;
  border-radius: 8px;
  text-decoration: none;
}

.hero a:hover {
  background:  var(--primary-color);
}

.products-section {
  padding: 40px 0;
}

.products-section h2 {
  text-align: center;
  margin-bottom: 25px;
}

.product-grid {
  display: grid;
  grid-template-columns: repeat(
    auto-fit,
    minmax(220px, 1fr)
  );
  gap: 25px;
}

.product-card {
  background: white;
  padding: 20px;
  border-radius: 12px;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.15);
  text-align: center;
  transition: 0.3s;
}

.product-card:hover {
  transform: translateY(-5px);
}

.product-card img {
  width: 100%;
  height: 200px;
  object-fit: cover;
  border-radius: 10px;
}
```

```
.product-card h3 {
  margin: 15px 0 8px;
}

.price {
  color: var(--secondary-color);
  font-weight: bold;
  font-size: 18px;
}

.product-card button,
.checkout-section button,
.remove-button {
  background: var(--secondary-color);
  color: white;
  border: none;
  padding: 10px 18px;
  border-radius: 8px;
  cursor: pointer;
  margin-top: 12px;
}

.product-card button:hover,
.checkout-section button:hover {
  background: var(--primary-color);
}

.cart-section {
  background: white;
  padding: 30px;
  margin: 40px 0;
  border-radius: 12px;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.12);
}

.cart-section h2 {
  text-align: center;
  margin-bottom: 25px;
}

.cart-item {
  border-bottom: 1px solid #ddd;
  padding: 20px 0;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.remove-button {
```

```
  background: #dc2626;
}

.remove-button:hover {
  background: #991b1b;
}

.grand-total {
  text-align: right;
  margin-top: 25px;
  color: var(--primary-color);
}

.empty-cart {
  text-align: center;
  font-size: 18px;
  padding: 20px;
}

.checkout-section {
  margin: 50px 0;
}

.checkout-section h2 {
  text-align: center;
  margin-bottom: 25px;
}

form {
  max-width: 650px;
  margin: auto;
  background: white;
  padding: 30px;
  border-radius: 12px;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.15);
}

.form-group {
  margin-bottom: 20px;
}

.form-group label {
  display: block;
  font-weight: bold;
  margin-bottom: 7px;
}

input[type="text"],
input[type="tel"],
textarea {
```

```
  width: 100%;
  padding: 11px;
  border: 1px solid #ccc;
  border-radius: 8px;
  font-size: 15px;
}

input:focus,
textarea:focus {
outline: none;
  border-color:  var(--secondary-color);
  box-shadow: 0 0 5px rgba(37, 99, 235, 0.4);
}

.payment-section {
  margin: 20px 0;
}

.payment-section h3 {
  margin-bottom: 10px;
}

.payment-section label {
  display: block;
  margin: 8px 0;
}

.payment-section input {
  margin-right: 8px;
}

.checkout-section button {
  width: 100%;
  font-size: 16px;
}

footer {
  background: var(--primary-color);
  color: white;
  text-align: center;
  padding: 25px;
  margin-top: 50px;
}

footer p {
  margin: 5px 0;
}

@media (max-width: 768px) {
  .navbar {
```

```
    flex-direction: column;
    gap: 15px;
  }

  nav {
    flex-direction: column;
    gap: 12px;
  }

  .hero h2 {
    font-size: 28px;
  }

  .product-grid {
    grid-template-columns:  1fr;
  }

  .cart-item {
    flex-direction: column;
    align-items: flex-start;
    gap: 15px;
  }

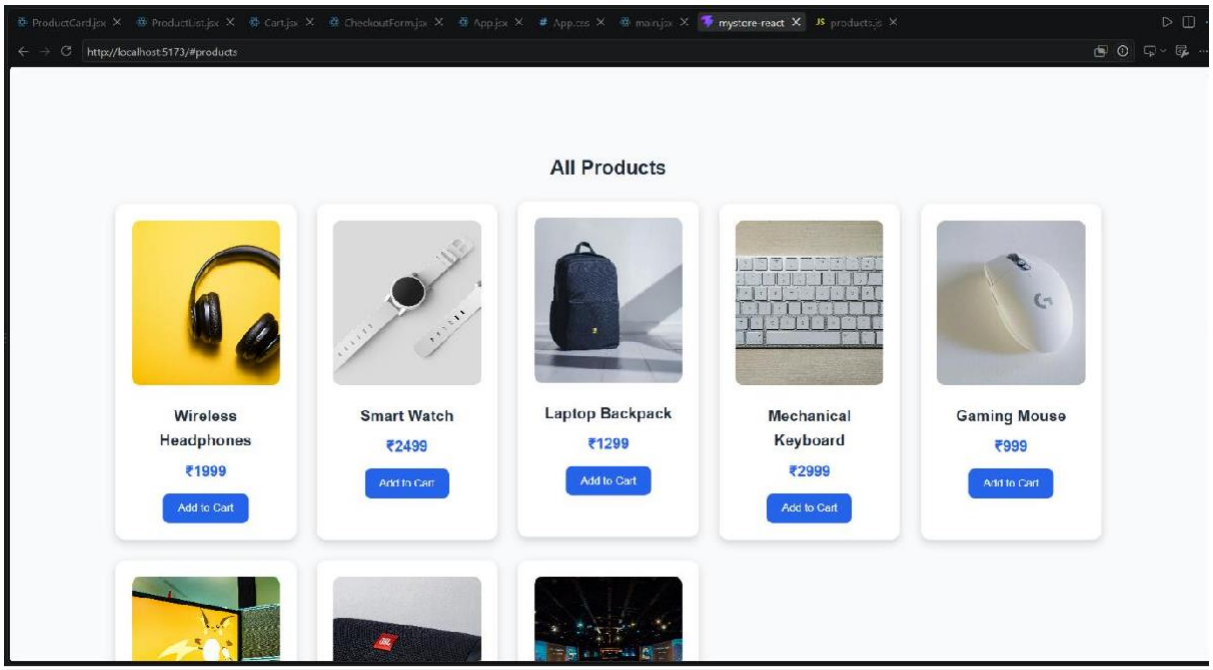
  .grand-total {
    text-align: left;
  }
}
```

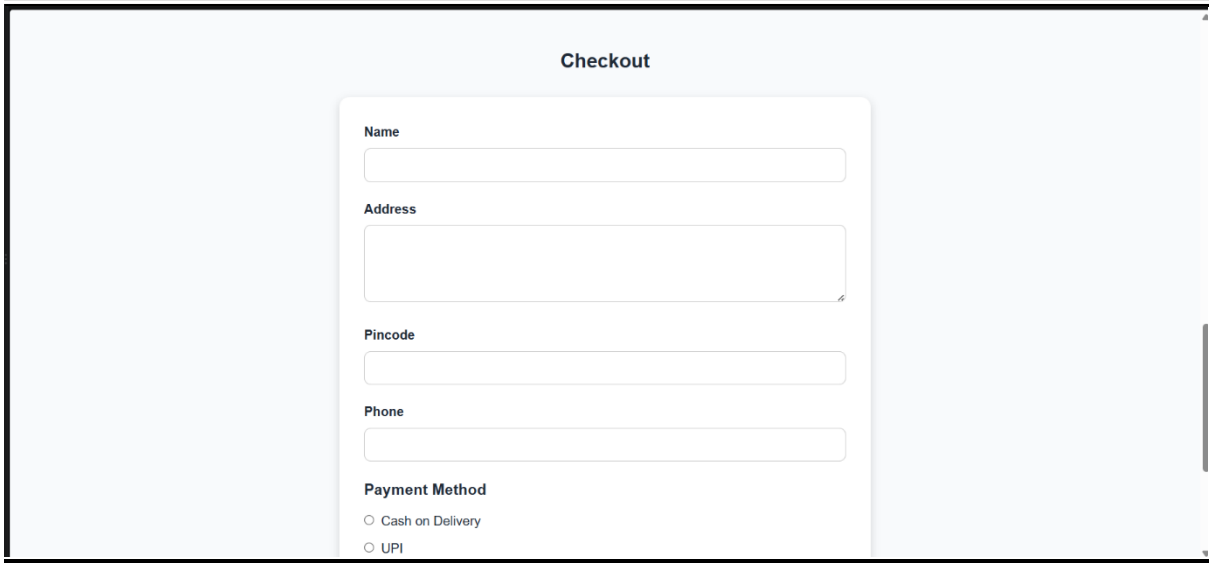
Products.js

```
const products = [
  {
    id: 1,
    name: "Wireless Headphones",
    price: 1999,
    image:   "https://images.unsplash.com/photo-1505740420928-5e560c06d30e"
  },
  {
    id: 2,
    name: "Smart Watch",
    price: 2499,
    image:   "https://images.unsplash.com/photo-1523275335684-37898b6baf30"
  },
  {
    id: 3,
    name: "Laptop Backpack",
    price: 1299,
    image:   "https://images.unsplash.com/photo-1553062407-98eeb64c6a62"
  },
  {
```

```
    id: 4,
    name: "Mechanical Keyboard",
    price: 2999,
    image:   "https://images.unsplash.com/photo-1587829741301-dc798b83add3"
  },
  {
    id: 5,
    name: "Gaming Mouse",
    price: 999,
    image:   "https://images.unsplash.com/photo-1527814050087-3793815479db"
  },
  {
    id: 6,
    name: "USB-C Hub",
    price: 1499,
    image:   "https://images.unsplash.com/photo-1625842268584-8f3296236761"
  },
  {
    id: 7,
    name: "Bluetooth Speaker",
    price: 1799,
    image:   "https://images.unsplash.com/photo-1608043152269-423dbba4e7e1"
  },
  {
    id: 8,
    name: "HD Webcam",
    price: 1599,
    image:   "https://images.unsplash.com/photo-1587825140708-dfaf72ae4b04"
  }
];

export default products;
```


ProductCard.jsx
ProductList.jsx
Cart.jsx
CheckoutForm.jsx
App.jsx
App.css
main.jsx
mystore-react
products.js
http://localhost:5173/#products
All Products
Wireless Headphones
₹1999
Add to Cart
Smart Watch
₹2499
Add to Cart
Laptop Backpack
₹1299
Add to Cart
Mechanical Keyboard
₹2999
Add to Cart
Gaming Mouse
₹999
Add to Cart


Checkout
Name
Address
Pincode
Phone
Payment Method
Cash on Delivery
UPI